



Comparison of differentiation of hiPSC-derived spinal neural progenitor cells in 2D and 3D cultures

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BACKGROUND

- SCI can have detrimental effects on autonomic, motor, and sensory function with minimal potential for recovery
- Cell replacement therapy (CRT) utilizes stem cell derived neural progenitor cells to restore lost connections and promote sensorimotor recovery and is a promising treatment for SCI
- Evidence has shown that 3D cultured grafts better mimic in vivo environment, increase cell type diversity, and increase capacity for host to graft interaction?
- We have developed protocols to differentiate human induced pluripotent stem cells (hiPSCs) into spinal neural progenitor cells (sNPCs) and 3D print them into scaffolds to create 3D cultures that recreate the environment of the spinal cord
- We report here our work assessing the effects of dimension and environment on spinal neural progenitor cell differentiation

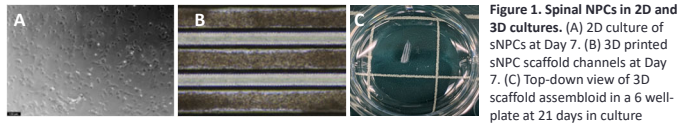


Figure 1. Spinal NPCs in 2D and 3D cultures. (A) 2D culture of sNPCs at Day 7. (B) 3D printed sNPC scaffold channels at Day 7. (C) Top-down view of 3D scaffold assembled in a 6 well-plate at 21 days in culture

METHODS

Establishment of Spinal Neural Progenitor Cells (sNPCs)

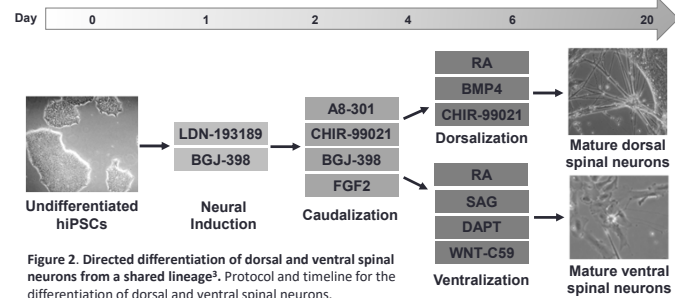


Figure 2. Directed differentiation of dorsal and ventral spinal neurons from a shared lineage¹. Protocol and timeline for the differentiation of dorsal and ventral spinal neurons.

2D and 3D Culture Conditions

- Following differentiation, sNPCs were cultured for 7 days, 27 days, and 77 days in both 2D and 3D conditions
- 2D cultures were plated on laminin and 3D cultures were printed in Matrigel bioink in silicone scaffolds
- Both cultures received the same neuronal maturation media and were maintained in an incubator at 37°C and 5% CO₂ for the duration of the experiment

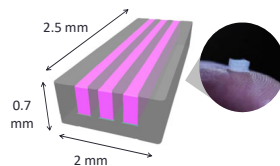


Figure 3. Construction of silicone scaffolds and sNPC bioink. Matrigel bioink (1:1) is printed at a density of 2x10⁸ cells/mL into three 200um scaffold channels

Molecular Techniques

- RT-qPCR
- Immunocytochemistry
- Patch clamp electrophysiology

RESULTS

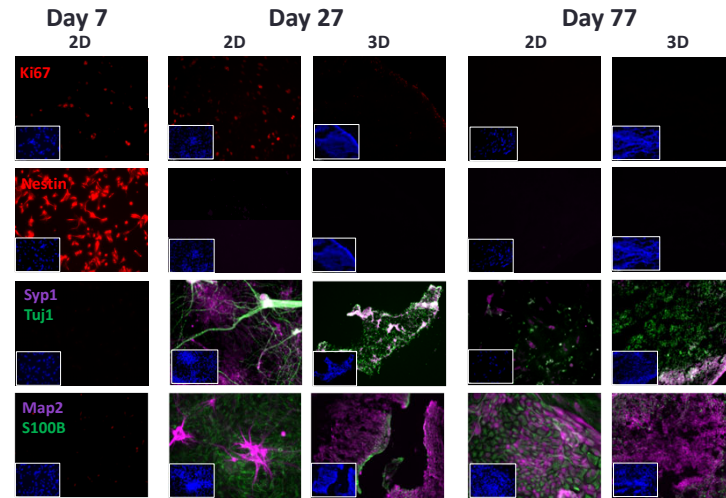


Figure 4. Representative images of 2D and 3D cultures at time points of 7, 27, and 77 days. Our protocol allows us to fully differentiate hiPSCs into sNPCs in 7 days. 3D cultures show decreased proliferation and increased neuronal expression, in relation to DAPI (in blue)

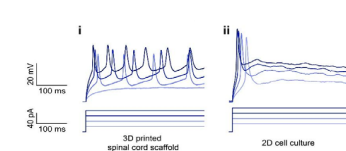


Figure 5. Electrophysiology of 2D and 3D cultures. Patch clamp recordings show (i) sNPCs printed in scaffolds exhibit enhanced maturity compared to (ii) 2D cultured cells

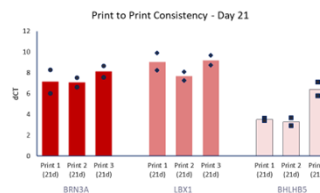


Figure 6. Print to print consistency. Relative gene expression normalized to actin across printing days

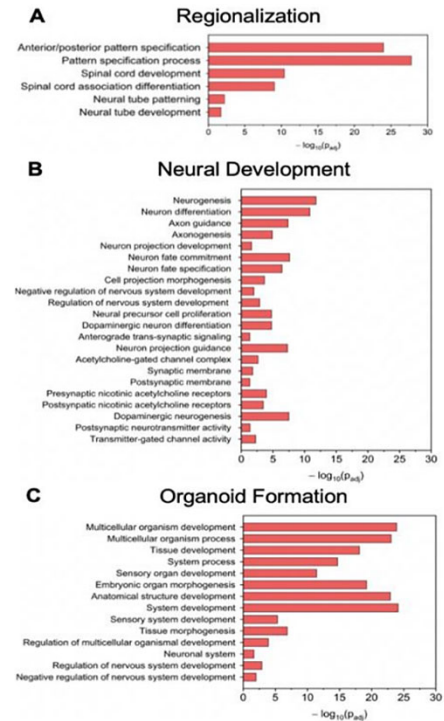


Figure 7. Gene ontology analysis reveals enrichment of spinal cord organogenesis and regionalization in 3D printed sNPCs compared to their 2D cultured equivalents

FUTURE DIRECTIONS

- To enhance the translation of CRT from bench to bedside, we propose a combinatorial approach that employs regionally specified transplants paired with the use of 3D printed bio-scaffolds to improve cell integration
- In the future, we aim to produce 3D cultures of cells with both dorsal and ventral regional specificity
- Future research will explore *in vitro* modeling of hiPSC-derived populations to study synaptic mechanisms of injury

References

1. Han, G., et al. (under review). 3D Printed Scaffolds Promote Enhanced Spinal Organoid Formation for Use in Spinal Cord Injury.
2. Duval, K., et al. (2017). Modeling Physiological Events in 2D vs. 3D Cell Culture.
3. Huntemer-Silveira, A., et al. (2023). Accelerated differentiation of human induced pluripotent stem cells into regionally specific dorsal and ventral spinal neural progenitor cells for application in spinal cord therapeutics.

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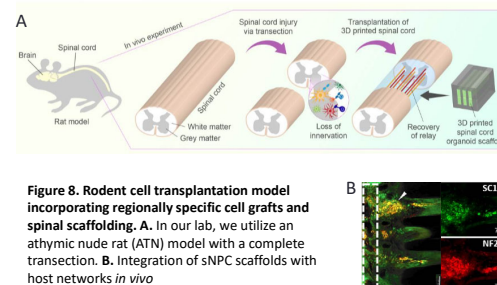


Figure 8. Rodent cell transplantation model incorporating regionally specific cell grafts and spinal scaffolding. A. In our lab, we utilize an athymic nude rat (ATN) model with a complete transection. B. Integration of sNPC scaffolds with host networks in vivo